

1. Administrative & Study Identification

Full Study Title: An Open-Label, Single-Dose Study to Evaluate the Pharmacokinetics of MK-8527 in Participants With Mild and Moderate Hepatic Impairment **Short Title/Acronym:** A Clinical Study of MK-8527 in Participants With Mild and Moderate Hepatic Impairment (MK-8527-015) **Protocol Number:** MK-8527-015 **NCT Number:** NCT07025551 **Sponsor Name:** Merck Sharp & Dohme LLC **Investigational Product:** MK-8527 (alimatravir) **Phase of Development:** Phase 1 **Medical Monitor:** Merck Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the pharmacokinetics (PK) of a single oral dose of MK-8527 in participants with mild and moderate hepatic impairment compared to healthy participants (AUC0-last, AUC0-inf, Cmax, Tmax). **Secondary Objectives:** To evaluate additional pharmacokinetic parameters (t1/2, CL/F, Vz/F) and assess the safety and tolerability of MK-8527 in this population.

2.2 Background & Rationale MK-8527 is a novel nucleoside reverse transcriptase translocation inhibitor (NRTTI) in development as a once-monthly oral tablet for HIV pre-exposure prophylaxis (PrEP). Because drugs are metabolized and cleared by the body, understanding MK-8527's pharmacokinetic profile in specific populations—such as individuals with mild or moderate hepatic impairment—is a critical step to inform clinical safety and establish appropriate dosing guidelines.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Parallel Assignment (Matched healthy control participants vs. hepatic impairment groups) **Blinding:** Open-label **Randomization:** Non-randomized

3.2 Planned Enrollment & Duration Target \$N\$: 18 (Actual Enrollment) **Number of Sites:** Global and Regional specialized PK sites **Estimated Study Start:** 19-Sep-2025 (Actual) **Estimated Primary Completion:** 13-Feb-2026 (Actual)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Continuous non-smoker or moderate smoker (≤ 10 cigarettes per day or equivalent) for at least 3 months prior to dosing. 2. Body mass index (BMI) ≥ 18.0 and ≤ 40.0 kg/m². 3. Participants with Mild HI (Group 1) and Moderate HI (Group 2): Has mild or moderate hepatic impairment with a diagnosis of chronic, stable, hepatic insufficiency featuring cirrhosis due to any etiology.

4.2 Exclusion Criteria 1. History of cancer (malignancy) or positive results for human immunodeficiency virus (HIV). 2. Positive for Hepatitis B surface antigen (HBsAg) or Hepatitis B core antibody (HBcAb), or positive for Hepatitis C Virus (HCV). 3. Has had major surgery and/or donated or lost significant volume of blood within 56 days prior to dosing.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Single oral dose of MK-8527 on Day 1. **Route of Administration:** Oral **Duration of Treatment:** 1 Day (single dose), followed by post-treatment monitoring for approximately 4 weeks.

5.2 Concomitant & Prohibited Therapy Permitted: Stable concomitant medications associated with the underlying hepatic condition, provided they are not strictly excluded by the protocol. **Prohibited:** Investigational drugs or medications known to significantly alter liver function or interact with the pharmacokinetic profile of MK-8527.

6. Study Timeline & Visit Schedule

Visit ID	Window	Key Activities/Procedures			
1	2-4 Weeks Prior	Informed Consent, Medical History, BMI, and Eligibility Labs	Baseline	Day 1	First Dose of MK-8527, initial safety and PK assessments
2	Over 4 Weeks	In-person visits for pre-specified plasma sample collection to determine PK	Interim		
3	Week 4 Post-Dose	Final PK and safety evaluation, AE/SAE monitoring	End of Study	Week 4	

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Pharmacokinetic parameters of MK-8527 including Area under the curve from time zero to last quantifiable concentration (AUC_{0-last}), AUC to infinity (AUC_{0-inf}), Maximum concentration (C_{max}), and Time to maximum concentration (T_{max}). **Measurement Method:** Plasma and blood samples analyzed at pre-specified timepoints using validated bioanalytical methods.

7.2 Secondary & Exploratory Endpoints 1. Additional Pharmacokinetic Parameters: Apparent terminal half-life ($t_{1/2}$), Apparent clearance (CL/F), and Apparent volume of distribution (Vz/F). 2. Safety and Tolerability: Incidence of any adverse events (AEs) or serious adverse events (SAEs) observed throughout the study.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection of any unfavorable and unintended sign (including abnormal lab findings), symptom, or disease temporally associated with the use of the investigational product. **Serious Adverse Events (SAEs):** Required continuous monitoring and expedited reporting per sponsor protocols. **Data Monitoring Committee (DMC):** Internal safety review managed by Merck Sharp & Dohme LLC.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Pharmacokinetic (PK) Analysis Set and Safety Data Set. **Sample Size Justification:** $N=18$; typical sample size adequate for Phase 1 single-dose PK studies comparing hepatic impairment populations to healthy controls. **Handling of Missing Data:** Participants with unquantifiable PK plasma samples treated per standard statistical procedures for Phase 1 pharmacokinetic modeling.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Standard site monitoring for Phase 1 clinical sites, focusing on proper plasma sampling, handling, and rigorous documentation of adverse events. **Audit Trail:** Secure eCRF locking and meticulous data validation for the PK bioanalytical assays.

11. Study Identifiers & Governance

Primary ID: MK-8527-015 **Registry Number:** NCT07025551 **Sponsor/Lead Organization:** Merck Sharp & Dohme LLC **Study Title:** A Clinical Study of MK-8527 in Participants With Mild and Moderate Hepatic Impairment (MK-8527-015) **Official Regulatory Title:** An Open-Label, Single-Dose Study to Evaluate the Pharmacokinetics of MK-8527 in Participants With Mild and Moderate Hepatic Impairment

12. Clinical Framework (Study Specific)

Primary Condition: Hepatic Impairment (Mild to Moderate) / Healthy Participants **Diagnosis Threshold:** Chronic, stable hepatic insufficiency with features of cirrhosis **Medical Methodology:** Pharmacokinetic (PK) tracking and safety profiling **Optimization Target:** Clarification of dosing and safety guidelines for PrEP populations with liver conditions

13. Study Procedures & Methodology

Management Pathway: Phase 1 Pharmacokinetics & Safety Monitoring Pathway **Education Protocol (HCP):** Site visit training focusing on precise PK timepoint blood collection and processing **Education Protocol (Patient):** Strict guidelines regarding fasting, diet, and smoking instructions around the dosing period **Quality Audit Mechanism:** Clinical site auditing for plasma storage conditions, AE tracking, and protocol adherence

14. Observation & Follow-Up Schedule

Total Duration: 4 weeks post-dose follow-up **Onsite Visit Cadence:** Multiple in-person visits required over 4 weeks for rigorous pre-specified plasma sample collection **Remote Monitoring Frequency:** N/A (Predominantly reliant on onsite clinic visits for PK labs) **Checklist Review Interval:** Continuous AE and SAE assessments during sample draws

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				